Sharp Gap-Free Bias Bounds for Cosine and Polynomial Singular-Value Filters

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Abstract

We analyze minimum-singular-value estimation by repeated spectral filtering on the normalized right Gram operator $G = (A/\alpha)^\dagger(A/\alpha)$. Rather than assume a gap to control eigenstate fidelity, we derive the exact worst-case bias of the filtered Gram mean over finite spectra with a given minimum and initial target mass. The extremizer has only two eigenvalues. This yields uniform gap-free bounds with asymptotically sharp constants: the cosine-Gram response $\cos(\beta G)$ has mean bias of order $t^{-1/2}$, whereas the distinct dilation response $\cos(\beta\sqrt{G})$ has order t^{-1} at t scalar iterations. The corresponding worst-case noiseless singular-value errors are of orders $t^{-1/4}$ and $t^{-1/2}$. Fixed-degree polynomial baselines attain the same rates. We also identify an exponential restart penalty caused by repeated global Taylor normalization and propagate success-block error through postselection and Bernoulli readout. Classical spectral audits check the sharp envelopes and the scaling penalty. These results refine the analysis of an existing proposal; they do not introduce a new quantum filtering primitive or establish a circuit-level advantage. Priority of the combined sharp analysis remains subject to further literature review.

1 Introduction

Extreme singular values arise in numerical linear algebra and in spectral questions connected to ground-state energy estimation. An earlier manuscript, *Quantum maximum/minimum singular-value estimation algorithms* (author and publication metadata not stated in the source file; hereafter QMSVE) proposes quantum power iteration for the largest singular value, an inverse-iteration baseline for the smallest, and a cosine-based triangular iteration that avoids explicitly applying a matrix inverse [1]. This report makes that proposal precise on a right-space operator, records the conditions needed for convergence, and checks a small set of finite-dimensional examples.

The main correction is about the operator being transformed. For a general rectangular matrix, applying $\cos(A)$ is not a valid singular-value prescription. We instead set $B = A/\alpha$ and $G = B^\dagger B$ on the right singular-vector space. The QMSVE cosine rule is then $C = \cos(\beta G)$, whose eigenvalues are $\cos(\beta\sigma_i^2/\alpha^2)$. This function is positive and decreasing on the normalized spectrum when $0 < \beta < \pi/2$. A Hermitian-dilation construction gives a different response, $\cos(\beta\sigma_i/\alpha)$; we keep it separate.

The main question is whether a useful filtered-mean error bound survives without a spectral-gap promise. We answer it for specified responses by an exact two-point worst-case reduction and sharp asymptotic constants (Section 5). A second outcome is a normalization audit and a finite-precision transfer bound (Section 6). The original geometric leakage bound is retained as background. These contributions improve the proposal's analysis rather than establish a new quantum algorithm.

2 Related work and positioning

Quantum linear algebra and power iteration. QSVT provides polynomial transformations of block-encoded singular values [2]. Quantum power iteration, inverse iteration, and QSVT-based power methods already exist [3, 4, 5]; neither iteration nor polynomial filtering is proposed as a new primitive here. Nghiem et al. explicitly analyze small success probabilities and modify the transformation to improve them. Their oracle construction differs from ours, so our scalar restart analysis is not a rebuttal of their complexity result.

Cosine and Gaussian filters. Ge, Tura, and Cirac already use powers of $\cos H$, response-ratio bounds, energy shifts, and amplitude amplification for ground-state preparation [8], notably Eqs. (7)–(22) of their arXiv version. Lu, Bañuls, and Cirac use cosine filtering for finite-energy observables, including reconstruction without physical filtered-state preparation [12]. He, Zhang, and Wang analyze a Gaussian filter and explicitly compare it to cosine filtering in Appendix C [13]. Thus the cosine-to-Gaussian local approximation is established background. A recent preprint also uses repeated power-cosine filtering with a single-ancilla time-evolution construction [14]. Its shifted/resonant response and implementation assumptions differ from the unshifted Gram response studied here; its performance claims are not adopted without audit.

Optimized projectors and input models. Lin and Tong optimize a polynomial projector for a known, isolated target [9], and study near-optimal preparation and energy estimation under additional information [10]. Dong, Lin, and Tong use controlled time evolution via QET-U [11]. Their gap, energy information, coherent depth, and amplification assumptions must be retained when making resource comparisons. Our maximization over spectra for a fixed response is distinct from minimax optimization over polynomials. Preserving unit response at the target is already explicit in Lin and Tong's construction; our normalization audit applies that principle to the Taylor proxy of the QMSVE draft.

Classical context and the contribution claimed here. Gap-independent accuracy guarantees also occur in classical power and Krylov methods [16]; gap-free approximation is not a new concept. The two-point extremization below is a simple linear-fractional argument [15]. The analytical contribution developed in this report is the exact fixed-response mean-bias envelope, its asymptotically sharp cosine-Gram versus dilation constants, and a joint audit of normalization and accepted-state perturbations. We did not identify this combined statement in the selected primary references; this bounded literature comparison does not establish priority over all classical or quantum filtering literature. The result is a modest analysis of specified estimators, not a new quantum algorithm or an advantage over optimized projectors.

The unpublished QMSVE source is treated as provenance of the proposal, with unconfirmed author and venue metadata [1]. The separate prior projects [6, 7] are not QMSVE implementations.

3 Model and algorithm

Let $A \in \mathbb{C}^{m\times n}$, with arbitrary positive m, n, and choose a known $\alpha \geq \|A\|_2 > 0$. Define

$$B = A/\alpha, \qquad G = B^\dagger B, \qquad 0 \leq \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_{n-1} \leq 1,$$

where $\lambda_i = \sigma_i(A)^2/\alpha^2$ are the eigenvalues of G in increasing order. Define the right minimum by $\sigma_0(A) = \min_{\|v\|=1} \|Av\|$. This includes right-kernel zero modes. For wide matrices it differs

from the smallest entry of the conventional thin SVD, which may be positive. Estimating the smallest nonzero singular value requires an additional promise or access to the kernel-orthogonal subspace. For wide matrices ($m < n$), the right Gram operator necessarily has a kernel, so the same distinction applies.

Let $|\psi\rangle = \sum_i c_i |v_i\rangle$ be a normalized input state in a right singular basis and set $p_i = |c_i|^2$. For $0 < \beta < \pi/2$, the ideal cosine-Gram filter is

$$C = \cos(\beta G), \qquad q_i = \cos(\beta \lambda_i) \in (0, 1].$$

The unnormalized state after t iterations is $C^t|\psi\rangle$, its ideal success probability is

$$Z_t = \|C^t|\psi\rangle\|^2 = \sum_i p_i q_i^{2t},$$

and the normalized accepted state's spectral weights are $p_i q_i^{2t}/Z_t$. This is the QMSVE rule applied to the right Gram operator, with normalization made explicit.

Polynomial implementation model. For an integer $m_0 \geq 0$, let

$$P_{m_0}(x) = \sum_{j=0}^{m_0} \frac{(-1)^j (\beta x)^{2j}}{(2j)!}, \qquad \delta_{m_0} = \frac{\beta^{2m_0+2}}{(2m_0+2)!}, \qquad \widetilde{P}_{m_0}(x) = \frac{P_{m_0}(x)}{1+\delta_{m_0}}.$$

The polynomial P_{m_0} is the Maclaurin polynomial of $\cos(\beta x)$ through degree $2m_0 + 1$ (the odd coefficient vanishes). Taylor's theorem and the bound on every derivative of cosine give $|P_{m_0}(x) - \cos(\beta x)| \leq \delta_{m_0}$ for $|x| \leq 1$. Hence $|\widetilde{P}_{m_0}(x)| \leq 1$ on $[-1, 1]$ and $\|\widetilde{P}_{m_0} - \cos(\beta\,\cdot)\|_\infty \leq 2\delta_{m_0}/(1 + \delta_{m_0})$; $\widetilde{P}_{m_0}$ is even of degree $2m_0$. It meets the boundedness and parity conditions for a real-polynomial Hermitian-QSVT response on a block-encoding of G; the completion and real-part extraction convention must still be fixed [2]. This establishes an admissible polynomial response, not a phase sequence, gate synthesis, or a compiled circuit. As a polynomial in the singular value σ/α, the response $\widetilde{P}_{m_0}((\sigma/\alpha)^2)$ has degree $4m_0$.

If a block-encoding of B is available, a product construction can encode $G = B^\dagger B$ using two calls to the B encoding, up to ancilla and reflection costs. We write d_G for the actual oracle-query cost of one implemented cosine-polynomial transform of G, and c_ψ for state preparation. Neither is numerically estimated here. A coherent implementation of t transforms has query depth proportional to td_G; a hard depth cap therefore constrains t as well as the polynomial degree. The normalization bound α and its acquisition cost are inputs, not free outputs of this method.

4 Finite-spectrum convergence and readout

Let $T = \{i : \lambda_i = \lambda_0\}$ be the minimum-eigenvalue eigenspace, with initial mass $a = \sum_{i \in T} p_i > 0$. Let $q_* = \cos(\beta\lambda_0)$ and suppose

$$q_i \leq r q_* \quad (i \notin T), \qquad 0 \leq r < 1.$$

For the cosine-Gram response, if an eigenvalue strictly larger than λ_0 exists, define $\lambda_+ = \min\{\lambda_i : \lambda_i > \lambda_0\}$; then $r = \cos(\beta\lambda_+)/\cos(\beta\lambda_0) < 1$. If no such eigenvalue exists, all spectral weight is already in the target subspace and $F_t = 1$.

Theorem 1 (Conditional spectral leakage)**.** *For every integer $t \geq 0$, let F_t be the probability mass of T in the normalized state $C^t|\psi\rangle/\sqrt{Z_t}$. Then*

$$1 - F_t \leq \frac{1-a}{a} r^{2t}.$$

If $\mu_t = \langle C^t\psi|G|C^t\psi\rangle/Z_t$, then

$$0 \leq \mu_t - \lambda_0 \leq (1-\lambda_0)(1-F_t) \leq \frac{1-a}{a} r^{2t}.$$

The proof is given in Appendix A. The inherited scalar Lean verification record is summarized in Appendix B; the new sharp-envelope and stability theorems are hand-proved here and are not Lean-verified. The identification of the finite weights with a matrix eigenbasis, the block-encoding implementation, and the readout interpretation are finite-dimensional mathematical derivations, not part of the scalar Lean theorem.

Finite-degree transfer condition. The leakage theorem also applies to a bounded polynomial response $\widetilde{q}_i = \widetilde{P}(\lambda_i)$, but boundedness alone does not ensure that it orders the target eigenvalue correctly. Let $q_{\text{out}} = \max_{i \notin T} q_i$ and let E be a uniform error bound $|\widetilde{P}(x) - \cos(\beta x)| \leq E$ on $[0,1]$. If $E < (q_* - q_{\text{out}})/2$, then $|\widetilde{q}_*| \geq q_* - E > 0$ and $|\widetilde{q}_i|/|\widetilde{q}_*| \leq (q_{\text{out}} + E)/(q_* - E) < 1$. The generic finite-weight bound therefore transfers with this ratio. This sufficient condition can be conservative and is not evaluated as a degree-optimal filter here.

Maximum-singular-value branch. For the power method, take response $q_i = \lambda_i$ and target the maximum eigenspace $T_{\max} = \{i : \lambda_i = \lambda_{\max}\}$. If its initial mass is $a_{\max} > 0$, $\lambda_{\max} > 0$, and $\lambda_- = \max\{\lambda_i : \lambda_i < \lambda_{\max}\}$ exists, the same finite-weight inequality applies with $q_* = \lambda_{\max}$ and $r = \lambda_-/\lambda_{\max}$. Its target leakage is at most $((1-a_{\max})/a_{\max})r^{2t}$, and its filtered mean obeys $0 \leq \lambda_{\max} - \mu_t \leq (1-F_t) \leq ((1-a_{\max})/a_{\max})r^{2t}$. The maximum singular value estimate is $\alpha\sqrt{\widehat{\mu}_t}$ with corresponding error at most $\alpha\sqrt{((1-a_{\max})/a_{\max})r^{2t} + \zeta}$. If $\lambda_{\max} < 1$, the unnormalized success probability is at most $\lambda_{\max}^{2t}$; this concentration step has a success cost as well.

Suppose a block-encoding of B is used to measure the success flag without postselecting on it. On an input $|\psi_t\rangle$, that flag is Bernoulli with mean $\langle\psi_t|B^\dagger B|\psi_t\rangle = \mu_t$. From M accepted filtered-state preparations, Hoeffding's inequality gives

$$|\widehat{\mu}_t - \mu_t| \leq \zeta \quad \text{with probability at least } 1-\delta \quad \text{if} \quad M \geq \frac{\log(2/\delta)}{2\zeta^2}.$$

Clip $\widehat{\mu}_t$ to $[0,1]$ and estimate $\sigma_0(A)$ by $\widehat{\sigma} = \alpha\sqrt{\widehat{\mu}_t}$. Then

$$|\widehat{\sigma} - \sigma_0(A)| \leq \alpha\sqrt{\frac{1-a}{a} r^{2t} + \zeta}.$$

This is an additive-error statement under an ideal Bernoulli readout model; it does not account for noise, state-preparation error, or an unknown α.

Success and cost accounting. For the exact ideal cosine response, the target component alone gives $Z_t \geq a q_*^{2t}$, while $q_i \leq q_*$ gives $Z_t \leq q_*^{2t}$. If each failed filter attempt is discarded and restarted from $|\psi\rangle$, the expected number of attempts per accepted copy is $1/Z_t$. For an implemented polynomial response $\widetilde{P}(G)$, the success probability is instead $\widetilde{Z}_t = \|\widetilde{P}(G)^t\psi\|^2$; a finite-degree cost

estimate must use $\widetilde{Z}_t$, not silently substitute ideal Z_t. A query proxy for M accepted readout samples is

$$M\left(\frac{c_\psi + td_G}{\widetilde{Z}_t} + d_B\right),$$

where d_B is the cost of one Bernoulli readout using the B block-encoding. This proxy is symbolic; finite-degree experiments below report only idealized spectral response probabilities and do not measure $\widetilde{Z}_t$ from a compiled circuit. Amplitude amplification, coherent recycling, and compiler-level gate costs are outside this accounting. In particular, when $\lambda_0 > 0$, $q_* < 1$ and the success probability can decay exponentially in t; a small conditional leakage does not by itself imply low total cost.

5 Sharp gap-free bounds for the filtered mean

Geometric convergence to an eigenspace requires spectral separation. Estimating an extremal eigenvalue by a filtered mean is a different task: nearby unwanted eigenvalues contribute little bias even when their probability mass remains large. We quantify this distinction without assuming a lower bound on the gap. Throughout this section, $0 < a < 1$, $t \geq 1$, and the target mass is exactly a. All results remain upper bounds when only a lower bound on a is available, by substituting that lower bound.

Theorem 2 (Exact two-point bias envelope)**.** *Let q be a continuous strictly positive response on $[\lambda_0, 1]$, $\lambda_0 < 1$, and set $R_t(x) = [q(x)/q(\lambda_0)]^{2t}$. Over all finite spectra with minimum λ_0 and all normalized input weights with mass a on that minimum, the largest possible filtered-mean bias is*

$$\mathcal{B}_t(q; a, \lambda_0) = \max_{\lambda_0 \leq x \leq 1} \frac{(1-a)(x-\lambda_0)R_t(x)}{a + (1-a)R_t(x)}. \tag{1}$$

The maximum is attained by a two-eigenvalue spectrum, with weights a and $1-a$. In particular, this is an exact worst-case envelope for a fixed filter, not an optimization over filters or a quantum-query lower bound.

Proof. Write the outside weights as $(1-a)\nu_i$, where $\sum_i \nu_i = 1$. If the right side of (1) is b, its pointwise definition gives $(1-a)(x-\lambda_0)R_t(x) \leq b[a + (1-a)R_t(x)]$. Averaging with ν_i and dividing by the positive denominator proves the upper bound. Conversely, put all outside mass on a maximizing x. The maximum is positive for some $x > \lambda_0$, so a maximizing spectrum can indeed have minimum mass exactly a. This is an elementary instance of linear-fractional extremization [15]; the reduction itself is not proposed as a new optimization principle. □

For comparison, let $q_{\mathrm{G}}(x) = \cos(\beta x)$ and $q_{\mathrm{D}}(x) = \cos(\beta\sqrt{x})$, with $0 < \beta < \pi/2$. The latter is the right-space response of the even cosine transform of a Hermitian dilation. Define, for $k = 1, 2$,

$$A_a = \frac{a}{1-a}, \qquad C_k(a) = \max_{y \geq 0} \frac{y^{1/k}}{1 + A_a e^y}, \qquad y_k = \frac{1}{k} + W_0\left(\frac{e^{-1/k}}{kA_a}\right). \tag{2}$$

Here W_0 is the nonnegative real branch of the Lambert function, and the maximum defining C_k is evaluated at y_k.

Theorem 3 (Uniform rates and sharp leading constants)**.** *Uniformly over* $0 \le \lambda_0 < 1$ *and over the finite spectra and weights in Theorem 2,*

$$\mu_t - \lambda_0 \le \begin{cases} C_2(a)/(\beta\sqrt{t}), & q = q_{\mathrm{G}}, \\ C_1(a)/(\beta^2 t), & q = q_{\mathrm{D}}. \end{cases} \tag{3}$$

For each fixed a, β*, the leading constants and powers of* t *are asymptotically attained by two-point spectra with* $\lambda_0 = 0$ *and* $x_t = (y_k/(t\beta^2))^{1/k}$*, using* $k = 2$ *for* G *and* $k = 1$ *for* D*. The worst-case noiseless errors of the estimator* $\alpha\sqrt{\mu_t}$ *therefore have rates* $t^{-1/4}$ *and* $t^{-1/2}$*, respectively. These statements concern scalar iterations, with no matched oracle-cost claim.*

Proof. Let $h = -\log q$. For the Gram response, $h'(x) = \beta\tan(\beta x) \ge \beta^2 x$, so, writing $d = x - \lambda_0$, $2t[h(x) - h(\lambda_0)] \ge t\beta^2(x^2 - \lambda_0^2) \ge t\beta^2 d^2$. For the dilation response, continuous extension at zero gives $h'(x) = \beta\tan(\beta\sqrt{x})/(2\sqrt{x}) \ge \beta^2/2$, hence the same expression is at least $t\beta^2 d$. The ratio in (1) increases with R_t. Substituting $R_t \le e^{-t\beta^2 d^k}$ and setting $y = t\beta^2 d^k$ proves (3). Differentiating the definition of C_k gives $(ky - 1)e^y = 1/A_a$, which has the unique solution (2). For the matching lower bound, take $\lambda_0 = 0$ and $x = x_t$. The expansion $2t\log\cos(\beta x_t^{k/2}) = -y_k + O(t^{-1})$ proves that the ratio of the attained bias to the upper bound tends to one. Finally, $|\sqrt{\mu_t} - \sqrt{\lambda_0}| \le \sqrt{\mu_t - \lambda_0}$, with equality at $\lambda_0 = 0$. □

Interpretation and limits. The extremizing spectrum changes with t. These rates are worst-case statements across spectra, not the asymptotic rate of a fixed gapped matrix. For any fixed positive minimum, $h'(\lambda_0) > 0$ and the same argument gives $\mathcal{B}_t \le C_1(a)/(2th'(\lambda_0))$ for these two responses. Conversely, there is no uniform gap-free improvement of target fidelity over a: an outside eigenvalue can approach λ_0 arbitrarily closely at each finite t. The distinction between mean bias and state fidelity is thus essential. The zero-minimum examples prove sharpness of the estimator's bias, not hardness of estimating a minimum already promised to be zero; strictly positive minima tending sufficiently quickly to zero give the same limiting constants by continuity.

The success probability still obeys $aq(\lambda_0)^{2t} \le Z_t \le q(\lambda_0)^{2t}$ for these decreasing responses. At the kernel boundary $Z_t \ge a$; at a fixed positive minimum it can be exponentially small. Thus a gap-free bias guarantee does not establish a uniform efficient preparation or estimation algorithm.

6 Normalization, polynomial baselines, and implementation error

A normalization-induced restart penalty. For the Taylor proxy used above, $P_{m_0}(0) = 1$ but $\widetilde{P}_{m_0}(0) = (1+\delta_{m_0})^{-1}$. A scalar rescaling leaves all conditional weights unchanged while multiplying the success probability by $(1 + \delta_{m_0})^{-2t}$. For example, with $\beta = 1.2$ and $m_0 = 1$, $\delta_{m_0} = 0.0864$; the penalty factor at $t = 128$ is 6.12×10^{-10}. Even on a kernel-bearing input, any fixed positive global margin destroys the ideal constant success floor as t grows. Keeping that factor bounded away from zero requires $t\log(1 + \delta_{m_0}) = O(1)$. Target normalization is already central to established polynomial projector constructions [9]; the observation here identifies its concrete consequence for this Taylor proxy.

Fixed-degree reference responses. For $0 < \beta < \sqrt{2}$ set $c = \beta^2/2 < 1$ and consider

$$Q_{\mathrm{G}}(x) = 1 - cx^2, \qquad Q_{\mathrm{D}}(x) = 1 - cx.$$

As polynomials in the normalized singular value s, these are $1 - cs^4$ and $1 - cs^2$, respectively; both are even and bounded between zero and one on $s \in [-1, 1]$, and both preserve the kernel

exactly at the response level. These elementary polynomial filters are comparison baselines, not new algorithmic primitives. The bounds and sharp constants of Theorem 3 hold for them as well: for $k = 2, 1$, $-\log(1 - cx^k)$ has derivative at least ckx^{k-1}, and its local expansion gives the same matching lower bound. This shows that the ideal cosine response is unnecessary for obtaining these scalar rates. No claim of degree-optimality follows, and compiled QSVT success blocks must realize the specified normalization; an additional factor per iteration would reintroduce the restart penalty.

Transfer to an approximate success block. Let C and $\widetilde{C}$ be contractions and $\|\widetilde{C} - C\| \le E$, with $u = C^t\psi$ and $Z_t = \|u\|^2 > 0$. Telescoping gives $\|\widetilde{C}^t\psi - u\| \le tE$ and $|\sqrt{\widetilde{Z}_t} - \sqrt{Z_t}| \le tE$. If $tE < \sqrt{Z_t}$, the implemented accepted state exists, its distance in vector norm from the ideal normalized state is at most $2tE/\sqrt{Z_t}$, and hence

$$|\widehat{\sigma} - \sigma_0| \le \alpha\sqrt{\mathcal{B}_t(q; a, \lambda_0) + 2tE/\sqrt{Z_t} + \zeta} \tag{4}$$

on the event that Bernoulli readout error is at most ζ. Indeed, normalization changes the vector error by at most a factor two, pure-state trace distance is no larger than that vector distance, and the expectation of $0 \le G \le I$ changes by at most trace distance. This is a conditional perturbation estimate, not a new stability principle; encoding noise, phase synthesis and normalization must all be included in E before (4) is used. At a kernel, $\sqrt{Z_t} \ge \sqrt{a}$ provides an explicit tolerance; for a positive minimum, conditioning may make this tolerance exponentially restrictive.

7 Numerical audit of the sharp bounds

The new deterministic script `experiments/qmsve_gapfree_envelope.py` evaluates the exact one-dimensional envelope and independently checks it against 1,000 random thirteen-level spectra, with seed 20261027, and dense 20,001-point grids. A stationary-point equation is used for the envelope; the Lambert expression for C_k is cross-checked by a scalar root equation. No tested mixture or grid value exceeded its predicted bound by more than 2×10^{-12}. These finite checks support the implementation and do not replace the proofs. The JSON records versions, base commit, source hash, parameters, maximizers, and success probabilities.

At $a = 0.4$, $C_2(a) = 0.360257$ and $C_1(a) = 0.378089$. All four responses reach at least 99.9% of their respective asymptotic bounds by $t = 65536$ for the tested overlaps $a = 0.1, 0.4, 0.9$. In the $a = 0.4$ polynomial-Gram extremizer at $t = 128$, the kernel-preserving response has success 0.6625; the globally scaled degree-two Taylor response has the identical posterior but success 4.05×10^{-10}. This comparison isolates the scaling penalty and does not compare full circuits.

8 Additional matrix-level checks

We ran a deterministic matrix-level simulation using exact classical eigendecompositions of seven right Gram matrices and fixed initial spectral weights. For each case, the input state is defined in the computed right-eigenvector basis; the calculation propagates the exact spectral weights rather than simulating a quantum state vector or circuit. The figure reports the exact $\cos(\beta G)$ response at $\beta = 1.2$. A separate Bernoulli sample uses seed 20261026 and 2,000 accepted spectral-model draws. The script and full JSON output are in `experiments/qmsve_matrix_level.py` and `results/qmsve_matrix_level.json`.

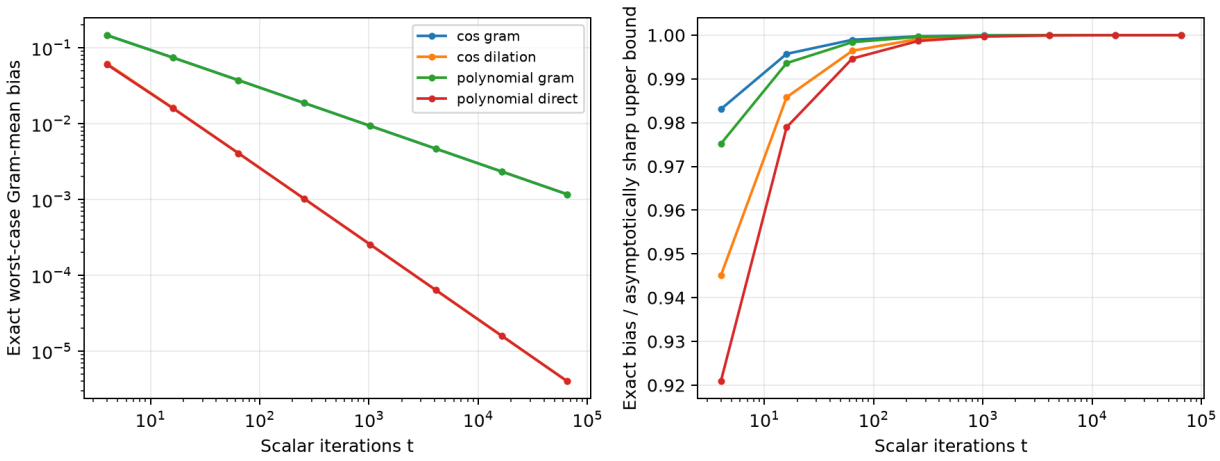


Figure 1: Exact worst-case mean bias at $a = 0.4$, $\beta = 1.2$, and its ratio to the sharp asymptotic upper bound. Each point uses its own two-level adversarial spectrum; it is not a trajectory on one fixed matrix. Scalar iterations are compared; oracle costs are not matched.

Case	$\sigma_{\min}$	$a_{\min}$	G^8 estimate	$\cos(\beta G)^8$ estimate / mass	$\cos(\beta\sqrt{G})^8$ estimate / mass
Square, separated	0.12	0.40	0.8600	0.2896 / 0.6791	0.1643 / 0.9417
Near-degenerate	0.20	0.30	0.8000	0.2076 / 0.4987	0.2027 / 0.5059
Degenerate minimum	0.20	0.40	0.7000	0.2713 / 0.9254	0.2041 / 0.9963
Zero target overlap	0.12	0	0.8600	0.4809 / 0	0.4803 / 0
Wide, full row rank	0	0.16	0.8000	0.2744 / 0.5666	0.1442 / 0.8715
Rank-deficient square	0	0.30	0.8000	0.2654 / 0.5714	0.1435 / 0.8719
Rotated tall, full column rank	0.15	0.35	0.6500	0.3065 / 0.8214	0.1630 / 0.9898

The G^8 column reports the estimated largest singular value; the cosine columns report the estimated smallest singular value and conditional minimum-eigenspace mass. For the square-gap case, G^8 has ideal success probability 0.00200, whereas the cosine-Gram response has success probability 0.5876. This distinction prevents a conditional estimate from being mistaken for a low-cost accepted sample.

For the square-gap example, finite-degree Taylor responses at $t = 8$ give:

Degree in G	Degree in σ/α	Uniform error bound	$\widehat{\sigma}$	Target mass	Ideal success probability
2	4	1.591×10^{-1}	0.2891	0.6801	0.1558
4	8	8.260×10^{-3}	0.2896	0.6791	0.5499
6	12	2.133×10^{-4}	0.2896	0.6791	0.5866
12	24	2.946×10^{-10}	0.2896	0.6791	0.5876

These are polynomial-response evaluations with a global analytic boundedness margin. They are not phase-angle synthesis results; the table reports the idealized spectral success probability, not a compiled-circuit success measurement.

At $t = 64$, the separated spectrum $(0.12, 0.48, 0.86)$ has minimum-eigenspace mass 0.994 and ideal filter success probability 0.395 under the original cosine-Gram response. The near-degenerate case $(0.20, 0.205, 0.80)$ reaches only mass 0.504 with success probability 0.514. A degenerate minimum with initial subspace mass 0.40 reaches mass greater than 0.999999; the method selects the eigenspace, not a unique vector. If the initial state has zero mass on the minimum eigenspace, the filter never creates that component: in the tested instance the estimate remains at 0.48 while the ideal success probability falls to 0.00282.

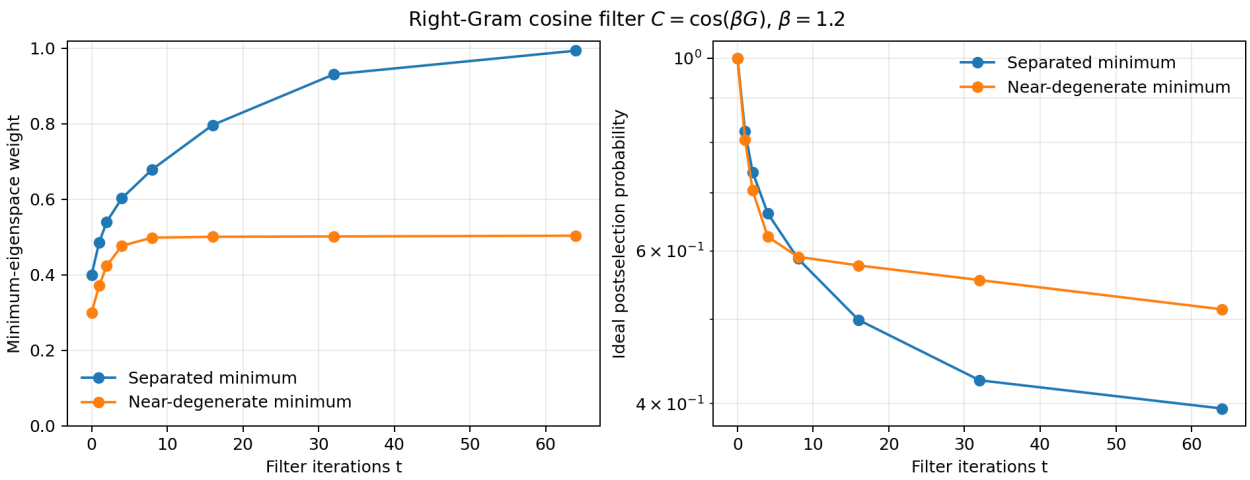


Figure 2: Exact matrix-level conditional target weight and ideal filter success probability. The separated case concentrates on the minimum eigenspace; the near-degenerate case does not do so over the displayed iteration range.

The matrix-level model also shows why a right-space definition matters. A wide full-row-rank matrix has a right-kernel eigenvalue zero in G; the method targets that kernel rather than the smallest nonzero singular value. The rank-deficient square example likewise converges toward its zero singular value. A normalization check rejects a matrix whose norm exceeds the declared $\alpha = 1$.

For the separated case at $t = 8$, the exact filtered Gram mean is 0.08385. The 2,000-shot Bernoulli estimate is 0.0780, within the stated 95% Hoeffding radius 0.03037. The expected number of fresh filter attempts for these accepted samples is approximately 3,404 under the ideal model. This is a scalar shot-noise check, not a circuit experiment.

The distinct dilation response $\cos(\beta\sigma)$ concentrates faster on the separated example (target mass 0.996 at $t = 16$ versus 0.797 for $\cos(\beta\sigma^2)$), but its success probability is lower (0.288 versus 0.500). For the near-degenerate example both responses remain close to 0.5 at $t = 16$. These numbers compare exact spectral responses only: their polynomial degrees, encodings, and compiled costs are not matched.

9 Discussion and limitations

The sharp envelope clarifies a limitation that eigenstate leakage alone conceals: a gap-free mean guarantee can hold even when no uniform target-fidelity guarantee is possible. Its exact leading constants expose the slower boundary behavior of the unshifted cosine-Gram response. The normalization example also shows why two identical conditional estimators can have exponentially different restart costs. These are estimator-specific analytical findings, not complexity lower bounds for general singular-value estimation.

The numerical audits use exact classical spectral information and ideal scalar success responses. They do not implement unknown singular-vector preparation, synthesize QSVT phases, or include physical noise. Positive target energy can cause exponential restart cost even when the scalar bias is small. Bernoulli/Hoeffding readout is only one readout model, and its precision dependence need not be optimal. A fair resource comparison must include state preparation, the normalization bound, block-encoding, polynomial degree, success amplification or restart, and readout. The power branch remains background; the new sharp results concern the minimum branch.

The two-point proof and polynomial baselines are elementary and overlap with established tools. The selected literature comparison supports a narrow analytical positioning, not a universal priority claim. Further review against classical weighted power-iteration analyses and additional cosine-filter papers is necessary before presenting the combined result as publishable novelty. Author metadata and original-source provenance also remain to be completed.

10 Conclusion

We derived an exact two-point envelope for filtered-mean bias and obtained sharp gap-free rates for specified cosine and polynomial responses on the right Gram spectrum. The analysis separates state fidelity, scalar estimation error, and acceptance cost. It identifies a concrete exponential penalty from repeated global normalization and supplies a postselection-aware perturbation bound. These results support a focused analytical paper draft; circuit realization, matched resource comparisons, and independent novelty assessment remain open.

A Proof of the finite-spectrum bounds

Write $Q_T = \sum_{i\in T} p_i q_*^{2t}$ for the unnormalized target weight and $Q_\perp = \sum_{i\notin T} p_i q_i^{2t}$. The assumed response bound implies

$$Q_\perp \le r^{2t} q_*^{2t} \sum_{i\notin T} p_i = (1-a) r^{2t} q_*^{2t}, \qquad Q_T = a q_*^{2t}.$$

Since $F_t = Q_T/(Q_T + Q_\perp)$, we have

$$1 - F_t = \frac{Q_\perp}{Q_T + Q_\perp} \le \frac{Q_\perp}{Q_T} \le \frac{1-a}{a} r^{2t}.$$

Every eigenvalue outside T lies in $[\lambda_0, 1]$, so the filtered mean satisfies

$$0 \le \mu_t - \lambda_0 = \sum_{i\notin T} \frac{p_i q_i^{2t}}{Z_t} (\lambda_i - \lambda_0) \le (1 - F_t)(1 - \lambda_0) \le \frac{1-a}{a} r^{2t}.$$

For a readout estimate with $|\widehat{\mu}_t - \mu_t| \le \zeta$, clipping to $[0,1]$ cannot increase its distance from $\lambda_0 \in [0,1]$. Thus $|\widehat{\mu}_t - \lambda_0| \le \mu_t - \lambda_0 + \zeta$. For nonnegative x, y, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$, giving the stated singular-value error bound after multiplication by α.

B Lean formalization status

The exact finite-weight postselection inequality and its posterior-moment consequence are checked in Lean 4. The unchanged formal source is included in the supplementary archive. Its inherited verification log records successful checks with Lean 4.35.0 and Mathlib v4.35.0 using `lake build QMSVE` and `lake env lean Audit.lean`, both with exit code 0. The matrix spectral decomposition, identification of filter weights with matrix eigencomponents, QSVT implementation, and Bernoulli concentration remain outside the scalar formalization.

C Source manuscript metadata

The QMSVE source used for this reassessment is the supplied manuscript titled *Quantum maximum/minimum singular-value estimation algorithms*. The copy examined does not state author, venue, or publication metadata, and labels the triangular iteration as provisional. The manuscript's author field is consequently left as a placeholder in the main file; no identity or affiliation is inferred.

D Reproducibility and verification scope

The supplementary source archive includes both experiment scripts, full JSON outputs and figures, dependency versions, and the unchanged scalar Lean project. The original Lean checker success is an inherited verification record, not a new CI run of this revision. The sharp-envelope, rate, polynomial-baseline and perturbation proofs have not been checked in Lean; finite numerical checks are reported separately. Full Lean source is distributed as `lean/QMSVE.lean`, rather than printed in the article.

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